OrientedFormer: An End-to-End Transformer-Based Oriented Object Detector in Remote Sensing Images

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***Abstract*— Oriented object detection in remote sensing images is a challenging task due to objects being distributed in multiorientation. Recently, end-to-end transformer-based methods have achieved success by eliminating the need for post-processing operators compared to traditional convolutional neural network (CNN)-based methods. However, directly extending transformers to oriented object detection presents three main issues: 1) objects rotate arbitrarily, necessitating the encoding of angles along with position and size; 2) the geometric relations of oriented objects are lacking in self-attention, due to the absence of interaction between content and positional queries; and 3) oriented objects cause misalignment, mainly between values and positional queries in cross-attention, making accurate classification and localization difficult. In this article, we propose an end-to-end transformer-based oriented object detector, consisting of three dedicated modules to address these issues. First, Gaussian positional encoding (PE) is proposed to encode the angle, position, and size of oriented boxes using Gaussian distributions. Second, Wasserstein self-attention is proposed to introduce geometric relations and facilitate interaction between content and positional queries by utilizing Gaussian Wasserstein distance scores. Third, oriented cross-attention is proposed to align values and positional queries by rotating sampling points around the positional query according to their angles. Experiments on six datasets DIOR-R, a series of DOTA, HRSC2016, and ICDAR2015 show the effectiveness of our approach. Compared with previous end-to-end detectors, the OrientedFormer gains 1.16 and 1.21 AP_{50} on DIOR-R and DOTA-v1.0, respectively, while reducing training epochs from 3× to 1×. The code is available at https://github.com/wokaikaixinxin/OrientedFormer.**

***Index Terms*— End-to-end detectors, oriented object detection, positional encoding (PE), remote sensing, transformer.**

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NOMENCLATURE

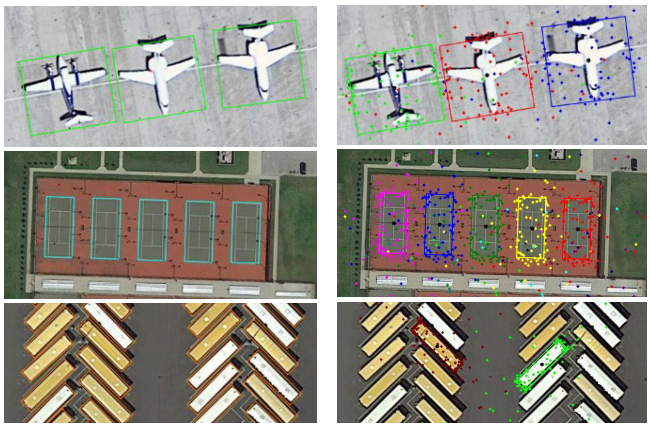
Notation	*Description*
(x, y, w, h, θ)	Oriented boxes.
(x, y, z, r, θ)	Oriented boxes.
$\{f^l\}_{l=1}^{L}$	Levels of feature map.
H_l	Height of feature map.
W_l	Width of feature map.
D	Channel of feature map.
N	Number of queries.
Q_c	Content query.
Q_p	Positional query.
$\varphi(\cdot)$	Positional encoding.
$\mathbf{x}$	Horizontal boxes.
T	Temperature of PE.
K, k	Hyperparameter of PE.
D'	Dimension of PE.
$\mathcal{N}(\cdot, \cdot)$	Gaussian distribution.
μ	Expectation.
$\mathbf{\Sigma}$	Variance.
$\mathbf{R}$	Rotation matrix.
$\mathbf{\Lambda}$	Diagonal matrix.
d_q	Dimension of self-attention.
G	Wasserstein distance score.
τ, ϵ	Coefficient of G.
$(\Delta x, \Delta y, \Delta z)$	Offset of sampling points.
g	Number of heads.
O	Number of sampling points.
$(\tilde{x}, \tilde{y}, \tilde{z})$	Sampling points.
P	Rotated sampling points.
s^l	Downsampling stride.
π_L	Scale-aware attention.
π_C	Channel-aware attention.
π_S	Spatial-aware attention.
V	Value of cross-attention.
V_{π_L}	Output of π_L.
V_{π_C}	Output of π_C.
η	Coefficient of π_L.
γ	Linear.
ρ	ReLU.
$\mathcal{L}$	Loss function.

I. INTRODUCTION

ORIENTED object detection is a fundamental task in the intersection of computer vision and remote sensing, as it aims to locate objects by a set of oriented boxes and categorize

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(a) (b)

Fig. 1. (a) Object instances distribute in remote sensing images with arbitrary orientation. Angles are used to characterize oriented objects, in addition to positions and sizes. (b) Visualization of sampling points of the oriented cross-attention for alignment.

them. Remote sensing images used in oriented object detection are photographs of target objects on the Earth's surface captured by satellites or other aerial platforms. Oriented object detection remains challenging due to objects being distributed with multiple orientations, dense arrangements, and varying scales, shown in Fig. 1(a). These characteristics of oriented objects make it difficult to localize and classify target objects accurately.

To detect objects accurately, oriented object detection methods [1], [2], [3] based on convolutional neural networks (CNNs) have made significant progress. Most of them are two-stage [4], [5], [6] or one-stage [7], [8], [9] detectors. Two-stage methods select foreground proposal boxes using a region proposal algorithm in the first stage and refine these proposals in the second stage. For example, RoI Transformer learns to transform horizontal anchors into oriented anchors, but horizontal anchors often misalign with the instance features. To address this issue, RRPN generates abundant oriented proposals using a rotation region proposal network (RPN). Oriented R-CNN introduces midpoint offset representation in an oriented RPN. Meanwhile, one-stage detectors directly predict the location and category of anchor boxes to avoid complex proposal generation. For example, R3Det directly obtains oriented proposals and aligns features through refinement modules. However, the above CNN-based methods rely on a one-to-many label assignment strategy and require complex hand-designed post-processing operators, such as non-maximum suppression (NMS).

DETR [10] first applies transformers [11] to horizontal object detection, and a series of related works [12], [13] have achieved promising performance. Inspired by them, some detectors adapt transformers for oriented object detection, typically following the encoder–decoder architecture. Compared to methods that adopt dense priors (e.g., boxes and points), transformer-based methods use a set of queries (e.g., content and positional queries) to represent object instances, which are usually updated layer by layer. The architecture of transformers mainly consists of three modules, positional encoding (PE), self-attention, and cross-attention. PE is used to learn the sequence ordering of input tokens. The self-attention handles pairwise interactions between queries and removes duplicate predictions. Cross-attention facilitates interactions between values and queries, enabling models to focus on related regions. Furthermore, transformer-based detectors benefit from one-to-one label assignment and an end-to-end framework, which eliminates the need for complex hand-designed processes.

However, extending the transformer framework to oriented object detection presents three main issues that need to be overcome.

1) *Objects rotate arbitrarily, requiring the encoding of angles* in addition to position and size. Angles are used to characterize oriented objects, which distinguishes them from horizontal boxes. Angles, position, and size are all necessary to represent oriented objects. Current transformer-based methods [14] encode only position and size using vanilla PE, but neglect angles. Additionally, we notice that the physical meanings and data ranges of angle θ, position (x, y), and size (w, h) are different. Without normalization, the value range for coordinates and sizes is bounded by the size of images, whereas angles are measured in radians, ranging from $[-\pi/2, \pi/2]$ for the rotation of oriented boxes. Therefore, vanilla PE is not suitable for oriented objects.
2) *The geometric relations of oriented objects are lacking in self-attention.* Content queries and positional queries represent semantic contextual features and spatial positions, respectively. Current transformer-based methods feed only content queries into self-attention, allowing them to interact with each other but failing to incorporate the geometric relation information provided by positional queries. Vanilla self-attention does not capture the geometric relationships between content queries.
3) *Oriented objects cause misalignment.* Objects rotate arbitrarily, while multiscale image features have a pyramidal structure. This typically leads to misalignment between values and positional queries in cross-attention, where values are extracted from image features and positional queries represent boxes of oriented objects. This makes it difficult to accurately classify and localize target objects.

In this article, we aim to alleviate the above issues for transformer-based oriented object detection. We propose an end-to-end transformer-based oriented object detection framework, called OrientedFormer. Our detector is equipped with three dedicated modules: Gaussian PE, Wasserstein self-attention, and oriented cross-attention. *First*, for the issue of encoding of angles, the Gaussian PE is built on Gaussian distribution transformed from oriented boxes. It unifies angle, position, and size into the same metric and effectively encodes the angles of boxes. *Second*, for the issue of lacking geometric relations, the Wasserstein self-attention enables content and positional queries to interact with each other. The geometric relation information is measured by Gaussian Wasserstein distance scores and provided by all positional queries. *Third*, for the issue of misalignment, the oriented

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cross-attention aligns values and positional queries. It rotates sparse sampling points around positional queries according to angles. The focused point regions distribute both inside and outside oriented boxes, which provides a wide range of contextual information, shown in Fig. 1(b). *Finally*, we conduct extensive experiments on various oriented object detection datasets of remote sensing images. All experimental results consistently demonstrate the effectiveness of OrientedFormer in improving accuracy. Furthermore, we validated the generalization of our method by conducting studies on scene text detection.

In general, the contribution of our methods is summarized in four folds as follows.

1) The Gaussian PE is proposed to encode the angle of oriented boxes in addition to position and size. It is constructed on Gaussian distributions, which unify angle, position, and size into the same metric.
2) The Wasserstein self-attention is proposed to introduce geometric relations into self-attention. This module utilizes Gaussian Wasserstein distance scores to measure the geometric relations between two different content queries.
3) To address the issue of misalignment, the oriented cross-attention is proposed to align values and positional queries by rotating a small set of sampling points around a positional query.
4) Extensive experiments on six datasets demonstrate the effectiveness of our approach. With the ResNet50 backbone, OrientedFormer achieves an AP_{50} of 67.28% on DIOR-R and 54.27% on DOTA-v2.0, respectively, establishing new state-of-the-art benchmarks.

II. Related Work

A. Oriented Object Detection in Remote Sensing

1) CNN Methods: These have achieved significant improvements in performance. Existing CNN-based oriented object detectors are mainly categorized into one-stage and two-stage methods. One-stage detectors predict the location and category of anchor boxes, which densely cover image feature maps, such as R3Det [15], S^2-Net [16], and PSC [17]. Anchor-free methods [9], [18] simplify the one-stage pipeline by replacing hand-crafted anchor boxes with prior points. One-stage methods rely on dense candidates, with each candidate supervised directly by classification and regression losses. During training, one-to-many assignment strategies based on pre-defined principles are used, such as whether the intersection over union (IoU) between candidates and ground truth boxes exceeds a threshold. Two-stage methods select foreground proposal boxes from dense region candidates in the first stage and localize and classify these proposal boxes in the second stage. The region proposal algorithm is used in the first stage to learn coarse proposal boxes, such as RPNs in Oriented R-CNN [4] and ReDet [5]. Similar label assignment strategies are required in two-stage methods. Some post-processing operators, for example, NMS, are needed to remove redundant prediction results during inference time both in one-stage and two-stage methods.

2) End-to-End Transformer-Based Methods: These are widely used in horizontal object detection [10], [12], which directly output the results without hand-crafted components. Zhou et al. [19] have extended them to oriented object detection. AO2-DETR [14] introduces an oriented boxes generation and refinement module for accurately oriented positional priors, building upon Deformable DETR [12]. ARS-DETR [20] improves on previous work by proposing rotated deformable attention, wherein sampling points are rotated based on angles for feature alignment. In addition, certain methods focus on improving object queries. PSD-SQ [21] represents object queries as point sets instead of oriented boxes to enable accurate instance feature sampling. D^2Q-DETR [22] designs dynamic queries that gradually reduce the number of object queries in the stacked decoder layers, aiming to better balance model precision and efficiency. Furthermore, several studies have concentrated on enhancing one-to-one label assignment. EMO2-DETR [23] observes and addresses the issue that one-to-one label assignment results in relative redundancy of object queries because objects are unevenly distributed in images. Different from existing approaches, to effectively encode oriented boxes, measure the geometric relations between content queries, and align values and positional queries, we propose the proposed Gaussian PE, Wasserstein self-attention, and oriented cross-attention.

B. Attention in Transformer-Based Object Detection

1) Self-Attention: Object queries are fed into self-attention and interact with each other to remove duplicate predictions [10]. Most transformer-based detectors adopt vanilla self-attention following DETR [10]. In vanilla self-attention, only content queries are used, and the geometric relations provided by positional queries are lost. As a result, vanilla self-attention does not account for geometric relations between content queries. In contrast to this approach, we introduce Gaussian Wasserstein distance scores into self-attention to measure the geometric relations between different content queries.

2) Cross-Attention: In cross-attention, image features serve as values and interact with queries. The vanilla cross-attention in DETR [10] only adopts a single feature map which is low efficiency. To accelerate the converging speed, deformable attention proposed in Deformable DETR [12] attends to a small set of sampling points around a reference. Features corresponding to these points learn to perform classification and regression. But, at the supervision of angles, sampling points learn to locate at special positions [14], for example, the catercorner and axis with boxes, which may be suboptimal. Anchor DETR [24] decouples attention into row and column attention and processes them successively. The row and column sequences of image features lack orientation and spatial information. The SMCA [25] proposes spatially modulated co-attention by constraining attention responses to be high near initially estimated bounding box locations. Dynamic DETR [26] designs RoI-based dynamic attention inspired by dynamic convolution [27] to assist the transformer in focusing on the region of interest. The above methods

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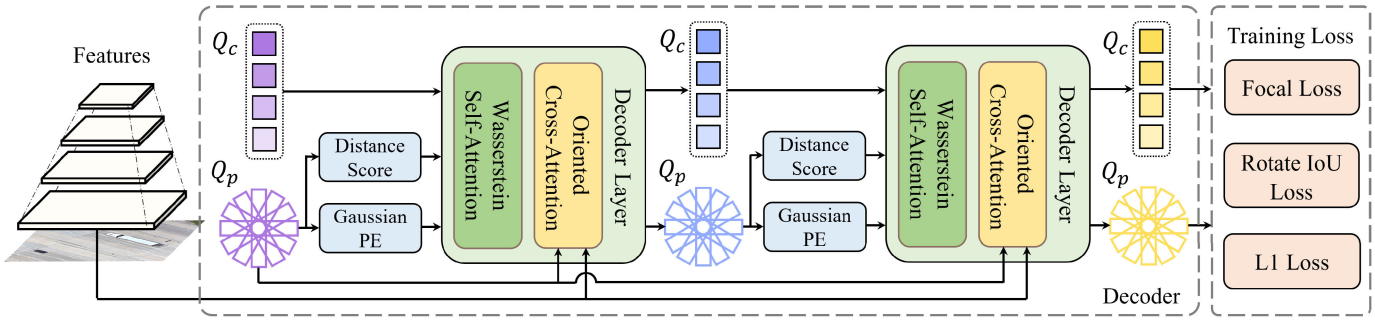


Fig. 2. Overall architecture of the OrientedFormer. Features are extracted from images. An object query is decomposed into a content query Q_c and a positional query Q_p. The Gaussian PE encodes positional queries. The Wasserstein self-attention measures the geometric relations between two different content queries by utilizing Wasserstein distance scores. The oriented cross-attention is proposed to align values and positional queries.

will cause misalignment when facing oriented boxes. Different from them, our oriented cross-attention aligns values and positional queries by rotating a small set of sampling points around a positional query.

C. Positional Encoding

PE is important for transformers to capture the sequence ordering of input tokens. It is first applied in the transformer [11] to inject information about the relative or absolute [11] position of tokens in a sequence for natural language processing. Since the transformer does not have convolution, it needs PE to learn the sequence ordering of tokens. The above methods are designed for 1-D word in sequence language models, and beyond that, PE is widely employed in computer vision. In object detection, DETR [10] employs learnable PE. The PE in DAB DETR [28] maps center coordinates (x, y) and size (w, h) of boxes to four vectors, respectively, and contacts them as final embeddings. The variant two-stage Deformable DETR [12] generates region proposals and then encodes them by sinusoidal absolute PE. The above methods only encode horizontal boxes. Different from them, our proposed Gaussian PE is constructed on Gaussian distribution, which is transformed from oriented boxes. It can encode the angles, position, and size of oriented boxes.

To facilitate clarity in the subsequent discussion, we present a summary of symbols along with their corresponding descriptions as utilized in this study, encapsulated in Nomenclature.

III. Method

In this article, we propose OrientedFormer, an end-to-end transformer-based oriented object detector in remote sensing images. In this part, we first introduce the overall architecture in Section III-A, and then illustrate object queries in Section III-B, Gaussian PE in Section III-C, Wasserstein self-attention in Section III-D, and oriented cross-attention in Section III-E, respectively. Lastly, label assignments and losses are introduced in Section III-F.

A. Overall Architecture

In general, the architecture of our OrientedFormer is composed of a backbone and a decoder, which follows the paradigm of end-to-end transformers, shown in Fig. 2. The encoder is not used and queries are initialized by enhancement method [29]. Multiscale image features $\{f^l\}_{l=1}^{L}$ are extracted by the backbone [30] and transformed in the same channel via the channel mapper [12], where $f^l \in \mathbb{R}^{D\times H_l\times W_l}$ and $l \in \{1, 2, \ldots, L\}$ denote a single-level feature and different scales, respectively. A single-level feature f^l is with channel D (256 by default), height H_l, and width W_l. The downsampling stride between two adjacent features is usually 2. Multiscale features and object queries are the inputs of the decoder. Following [10] and [12], we sequentially use our proposed self-attention, cross-attention, and feedforward-feed network (FFN) in the decoder. In self-attention, object queries interact with each other, while in cross-attention, sampled features as values further interact with queries. Through FFN, updated queries and detection results are produced. During training times, predictions are supervised by classification and regression losses.

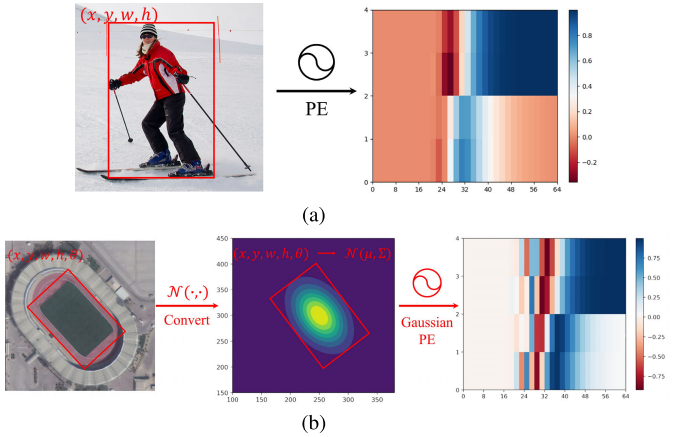


Fig. 3. An example of Gaussian PE. (a) PE in Deformable DETR. (b) Gaussian PE.

B. Object Queries

Object queries are one of the inputs of the decoder for representations of object instances. Content queries $Q_c \in \mathbb{R}^{N\times D}$ and positional queries $Q_p \in \mathbb{R}^{N\times 5}$ are used to learn contextual image information and positions of objects, respectively, where N denotes the number of object queries and D (256 by default) denotes channel dimension. These two

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kinds of queries disentangle the classification and localization of objects.

C. Gaussian PE

The modern PE is exclusively employed for horizontal boxes and inaccurately encodes angles of oriented objects. To address the limitations, we propose Gaussian PE, which can encode angle, position, and size uniformly, as shown in Fig. 3.

1) Preliminaries of PE: In the decoder, PE transforms positional queries into sinusoidal embeddings, and then content queries are trained jointly with positional embeddings. We review modern PE in object detection first. In many common horizontal object detectors [12], [13], PE applied to queries can be formulated as

$$\varphi(\mathbf{x}) = \left[\sin(\mathbf{x}), \cos(\mathbf{x}), \ldots, \sin\left(\frac{\mathbf{x}}{T^{\frac{2K}{D'}}}\right), \cos\left(\frac{\mathbf{x}}{T^{\frac{2K}{D'}}}\right)\right]^{\top}. \quad (1)$$

This is the concatenation of sine and cosine values of each dimension of the horizontal boxes $\mathbf{x}$ (x, y, w, h) scaled by $1/T^{(2k/D')}$, where T, $k \in \{0, 1, \ldots, K\}$, and D' are temperature, hyperparameter, and dimension, respectively.

2) Gaussian Distribution of Oriented Boxes: To unify angle, position, and size into the same metric, we convert an oriented box into a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\begin{aligned}\boldsymbol{\mu} &= (x, y)\\ \boldsymbol{\Sigma} &= \mathbf{R}\boldsymbol{\Lambda}\mathbf{R}^{\top}\\ &= \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\frac{w^2}{4} & 0\\ 0 & \frac{h^2}{4}\end{pmatrix}\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\end{aligned} \quad (2)$$

where $\mathbf{R}$ is the 2-D rotation matrix, and $\boldsymbol{\Lambda}$ is the diagonal matrix of eigenvalues.

3) Gaussian PE: The proposed Gaussian PE is the expectation of oriented boxes distributed according to the aforementioned Gaussian. The original PE in (1) can be rewritten as

$$\varphi(\mathbf{x})_{2i} = \sin\left(T^{-\frac{2i}{D'}} \cdot \mathbf{x}\right), \quad \varphi(\mathbf{x})_{2i+1} = \cos\left(T^{-\frac{2i}{D'}} \cdot \mathbf{x}\right) \quad (3)$$

where the superscript $2i$ and $2i+1$ denote the indices in the encoded vectors. This reparameterization makes it possible to derive a closed form for Gaussian PE. There are two mathematical facts: the expectation of a linear transformation of random variables is a linear transformation of the random variables' expectation, and the variance of a linear transformation of random variables is the product of variance and the square of coefficient. According to these properties, we can calculate the mean and covariance of Gaussian distributions of oriented boxes after they are lifted for PE

$$\boldsymbol{\mu}_{\varphi} = T^{-\frac{2i}{D'}} \cdot \boldsymbol{\mu}, \quad \boldsymbol{\Sigma}_{\varphi} = \left(T^{-\frac{2i}{D'}}\right)^2 \cdot \boldsymbol{\Sigma}. \quad (4)$$

The last step of our Gaussian PE is calculating expectations over the lifted multivariate Gaussian in (4), which is modulated by sinusoidal and cosine functions. There is another mathematic fact that, if x distributes in Gaussian with mean μ and variance σ, the expectation values $\mathrm{E}[\sin(x)]$ and $\mathrm{E}[\cos(x)]$ are [31]

$$\begin{aligned}\mathrm{E}_{x\sim\mathcal{N}(\mu,\sigma^2)}[\sin(x)] &= \sin(\mu)\exp\left(-(1/2)\sigma^2\right)\\ \mathrm{E}_{x\sim\mathcal{N}(\mu,\sigma^2)}[\cos(x)] &= \cos(\mu)\exp\left(-(1/2)\sigma^2\right)\end{aligned} \quad (5)$$

as we can see, the mathematical expectations $\mathrm{E}[\sin(x)]$ and $\mathrm{E}[\cos(x)]$ are the $\sin(\cdot)$ and $\cos(\cdot)$ of mean μ attenuated by $\exp(\cdot)$ of the variance σ. With the property, the proposed Gaussian PE is calculated as

$$\begin{aligned}\varphi(\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \mathrm{E}_{\mathbf{x}\sim\mathcal{N}(\boldsymbol{\mu}_{\varphi}, \boldsymbol{\Sigma}_{\varphi})}[\varphi(\mathbf{x})]\\ &= \begin{bmatrix}\sin(\boldsymbol{\mu}_{\varphi}) \circ \exp(-(1/2)\mathrm{diag}(\boldsymbol{\Sigma}_{\varphi}))\\ \cos(\boldsymbol{\mu}_{\varphi}) \circ \exp(-(1/2)\mathrm{diag}(\boldsymbol{\Sigma}_{\varphi}))\end{bmatrix}\end{aligned} \quad (6)$$

where $\circ$ denotes the Hadamard product. Since PE encodes each dimension of boxes $\mathbf{x}$ independently, the Gaussian PE depends on only the marginal distribution of $\varphi(\mathbf{x})$. Thus, only the diagonal of the covariance matrix $\boldsymbol{\Sigma}_{\varphi}$ is required

$$\mathrm{diag}(\boldsymbol{\Sigma}_{\varphi}) = \left[\mathrm{diag}(\boldsymbol{\Sigma}), \ldots, \left(\frac{1}{T^{2K/D'}}\right)^2 \mathrm{diag}(\boldsymbol{\Sigma})\right]^{\top}. \quad (7)$$

These diagonals can be easily obtained from the variance $\boldsymbol{\Sigma}$ of Gaussian distribution of oriented boxes in (2).

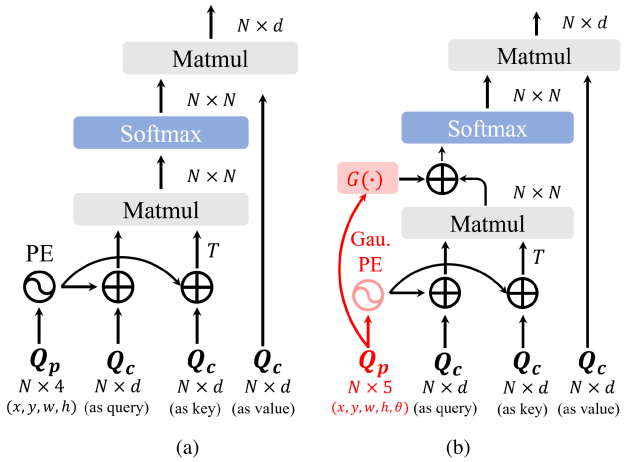


Fig. 4. Self-attention in the decoder. (a) Vanilla self-attention. (b) Wasserstein self-attention.

D. Multihead Wasserstein Self-Attention in Decoder

The vanilla multihead self-attention [10] used between content queries does not account for geometric relation information. To address this, we propose Wasserstein self-attention, which introduces geometric relations into self-attention mechanisms and can effectively suppress redundant detections [10], as shown in Fig. 4.

1) Wasserstein Self-Attention: We introduce Gaussian Wasserstein distance scores into the self-attention. It can measure the geometric relations between two different queries and assist self-attention focus on important areas, which is the main difference between Wasserstein self-attention and other vanilla self-attention. Given any two positional queries $\mathbf{x}_1$, $\mathbf{x}_2$ with Gaussian distributions $\mathbf{x}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathbf{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ in (2), the Wasserstein distance is calculated as

$$d_{ij}^2 = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|_2^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2\left(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2}\right)^{1/2}\right) \quad (8)$$

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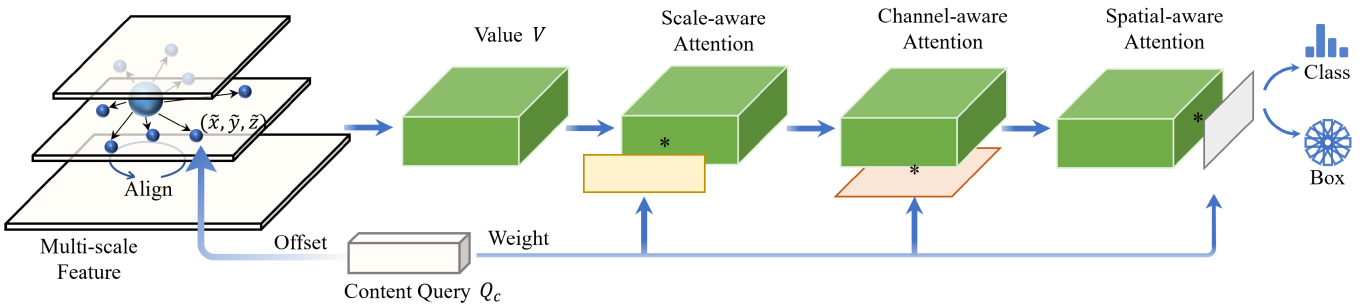


Fig. 5. Oriented cross-attention. It attends to sparse sampling points $(\tilde{x}, \tilde{y}, \tilde{z})$ around the center of a positional query. Sampling points are rotated according to angles for alignment. Values V are interpolated by sampling points and multiscale features. We deploy attention mechanisms separately on each particular dimension of values, that is, scale-aware, channel-aware, and spatial-aware.

where $i, j \in \{1, 2, \ldots, N\}$ and $i,\ j$ denote two Gaussian distributions of any two positional queries. The distance only satisfies ≥ 0. Thus, we further rescale it and finally get the Gaussian Wasserstein distance scores

$$G_{ij} = \log\left(\frac{1}{\tau + d_{ij}} + \epsilon\right) \tag{9}$$

where $\tau = 1$, $\epsilon = 10^{-7}$, and $i, j \in \{1, 2, \ldots, N\}$. The $G_{ij} = 0$ stands for the positional queries i and j coinciding, and $G_{ij} = \log \epsilon \ll 0$ indicates that the two positional queries i and j are far away from each other. The proposed multihead Wasserstein self-attention combines Gaussian PE and Gaussian Wasserstein distance scores

$$\text{WAttn}(Q_c, \varphi, G) = \text{Softmax}((Q_c + \varphi(\boldsymbol{\mu}, \boldsymbol{\Sigma}))(Q_c + \varphi(\boldsymbol{\mu}, \boldsymbol{\Sigma}))^\top / \sqrt{d_q} + G) Q_c. \tag{10}$$

2) Complexity of Wasserstein Self-Attention: The complexity of Wasserstein self-attention is $\mathcal{O}(D^2N + DN^2)$, on the same order of magnitude as other methods [10], [20]. The Gaussian Wasserstein distance scores and Gaussian PE do not impose additional computational burden.

E. Multihead Oriented Cross-Attention in Decoder

We propose oriented cross-attention for the issue of misalignment, shown in Fig. 5. The inputs of oriented cross-attention contain multiscale image features $\{f^l\}_{l=1}^L$, content queries $Q_c \in \mathbb{R}^{N \times D}$, and positional queries $Q_p \in \mathbb{R}^{N \times 5}$, where f^l and l denote a single-level feature and different scales, respectively. Oriented cross-attention can be analyzed from three perspectives: 1) positional queries are transformed into another type (x, y, z, r, θ), which provide virtual 3-D coordinates; 2) values are sampled from image features, and sampling points are rotated according to angles for alignment; and 3) the proposed cross-attention can be decoupled into three different attention mechanisms, each focusing on a different perspective: scale-aware attention, spatial-aware attention, and channel-aware attention.

The differences between oriented cross-attention and deformable attention are as follows: 1) our attention rotates sampling points according to angles for alignment, while deformable attention does not; 2) our attention focuses on three perspectives: scale-aware, spatial-aware, and channel-aware attention, while deformable attention only emphasizes channel-awareness; 3) our attention utilizes learnable positional queries, whereas deformable attention employs a set of fixed mesh grids as reference points; and 4) sampling points in our attention are distributed in a virtual 3-D feature space, while in deformable attention, they are confined to a 2-D plane.

1) Coordinates of Positional Queries: We transform positional queries into another type $(x, y, z, r, \theta) \in \mathbb{R}^5$, where (x, y) indicates coordinates of the center point, (z, r) denotes the logarithm of scale and aspect ratio, and θ represents the angle. The only difference between (x, y, z, r, θ) and the common five-parameter representation (x, y, w, h, θ) is (w, h), and the conversion is as follows:

$$z = \log_2 \sqrt{wh}, \quad r = \log_2 \frac{h}{w} \tag{11}$$

where (w, h) denotes the width and height, respectively, of oriented boxes. The purpose is that (x, y, z) represents 3-D coordinates in the virtual 3-D feature space. Benefiting from this, it is easy to achieve scale, spatial, and channel attention.

2) Calculation of Values and Feature Alignment: We sample features as values. Sampling points can be obtained from offsets of the center (x, y, z) of a positional query $Q_{p_i}(x, y, z, r, \theta)$. The offsets $(\Delta x, \Delta y, \Delta z)$ are calculated as

$$(\Delta x, \Delta y, \Delta z) = \text{Linear}(Q_c). \tag{12}$$

These offsets are transformed into sampling points

$$\begin{cases} \tilde{x} = x + \Delta x \cdot 2^{z-r} \\ \tilde{y} = y + \Delta y \cdot 2^{z+r} \\ \tilde{z} = z + \Delta z. \end{cases} \tag{13}$$

Like other popular cross-attention, we also introduce multiple heads in oriented cross-attention. Thus, the number of sampling points around a position query is $g \cdot O$, where g and O denote the number of heads and sampling points, respectively.

Because objects in remote sensing images are oriented, we need to align the sampling points according to angles θ

$$P = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix}. \tag{14}$$

The $\tilde{z}$ does not participate in rotating, because oriented boxes only rotate in 2-D planes. Our task is 2-D oriented object detection, which cannot obtain real depth information similar to other 3-D tasks. Thus, we need to rescale sampling points

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to adapt to different levels of features. The parameter $\tilde{z}$ is not directly involved in feature sampling but will be transformed into attention weights in scale-aware attention, which will be elaborated below. Given the aligned sampling points P, we rescale them first, and then sample values V by bilinear interpolation in every level of features $\{f^l\}_{l=1}^{L}$

$$V^l = \text{interpolation}(f^l, P/s^l) \tag{15}$$

where s^l denotes the downsampling stride of each level of features, and $l \in \{1, 2, \ldots, L\}$. The aligned sampling points are not strictly restricted to boxes. The values V^l now are of the shape $\mathbb{R}^{N\times g\times O\times (D/g)}$, because the multihead mechanism is used, and the number of sampling points around a position query is $g \cdot O$.

3) Scale-Aware Attention: We introduce scale-aware attention, which dynamically fuses features of different scales

$$\pi_L(Q_c, V) = \sum_{l=1}^{L} \text{Sigmoid}(-(\tilde{z} - \log_2(s^l))^2/\eta) \cdot V^l \tag{16}$$

where s^l denotes the downsampling stride of each level of features, and $l \in \{1, 2, \ldots, L\}$. The η is the softing coefficient for the weight over different scales and we keep $\eta = 2$ in the work. We denote the output of scale-aware attention as V_{π_L}. The V_{π_L} is of the shape $\mathbb{R}^{N\times g\times O\times (D/g)}$ and continuously used in the channel-aware attention.

4) Channel-Aware Attention: To pay attention to channel dimensions, we introduce channel-aware attention. Given the output of scale-aware attention V_{π_L} and content queries Q_c, the channel-aware attention is calculated as

$$\pi_C(Q_c, V_{\pi_L}) = \text{Sigmoid}(\gamma_1(\rho(\gamma_1(Q_c)))) \cdot V_{\pi_L} \tag{17}$$

where γ_1 and ρ are linear and ReLU operator, respectively. The weights for the channel are transformed from content queries by linear and ReLU operators, and have the shape of $\mathbb{R}^{N\times g\times 1\times (D/g)}$. We denote the output of channel-aware attention as V_{π_C}. The V_{π_C} is of the shape $\mathbb{R}^{N\times g\times O\times (D/g)}$ and continuously used in the spatial-aware attention.

5) Spatial-Aware Attention: To focus on spatial dimensions, we introduce spatial-aware attention. Given the output of channel-aware attention V_{π_C} and content queries Q_c, the spatial-aware attention is calculated as

$$\pi_S(Q_c, V_{\pi_C}) = \text{Sigmoid}(\gamma_2(\rho(\gamma_2(Q_c)))) \cdot V_{\pi_C} \tag{18}$$

where γ_2 is the linear operator. The weights for spatial are also transformed from content queries by Linear and ReLU operators and have the shape of $\mathbb{R}^{N\times g\times O\times 1}$. The output of spatial-aware attention is of the shape $\mathbb{R}^{N\times g\times O\times (D/g)}$. The output is further flattened and transformed to $\mathbb{R}^{N\times D}$ by a linear layer as the final output of oriented cross-attention to add back to the content queries.

F. Label Assignment and Loss

In oriented object detection tasks, there are two subtasks, classification for categories and regression for positions of objects. At the stage of label assignment, one-to-one Hungarian matching [32] is used. The losses consist of the focal loss [33] for classification, and L_1 loss and rotate IoU loss [34] for regression

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{L_1}\mathcal{L}_1 + \lambda_{\text{iou}}\mathcal{L}_{\text{iou}} \tag{19}$$

where λ_{cls}, λ_{L_1}, and λ_{iou} are coefficient of corresponding losses. The losses were applied at the detection results of all the decoder layers for training.

Algorithm 1 Training Procedure

Input: Images; Real Boxes; Real Classes.
Output: Loss $\mathcal{L}$.

Decoder_Layer $\leftarrow$ 2; Feature Layer $L \leftarrow$ 5; $\lambda_{cls} \leftarrow$ 2; $\lambda_{L_1} \leftarrow$ 2; $\lambda_{iou} \leftarrow$ 5
$\{f^l\}_{l=1}^{L}$ = Backbone_with_FPN(Images)
Q_c, Q_p = Initialize_Query($\{f^l\}_{l=1}^{L}$)
for $i = 0$ to Decoder_Layer **do**
  φ = Gaussian PE(Q_p)
  G = Gaussian Wasserstein distance score(Q_p)
  Q_c = Wasserstein Self-attention(Q_c, φ, G)
  Q_c = FFN(Oriented Cross-attention($Q_c, Q_p, \{f^l\}_{l=1}^{L}$))
  $Q_p = Q_p +$ `Linear1` (Q_c)
  class_score = `Linear2` (Q_c)
  Q_p^i, class_score^i = Hungarian assignment({class_score, Real Classes}, {Q_p, Real Boxes})
  $\mathcal{L}_{cls}^i, \mathcal{L}_1^i, \mathcal{L}_{iou}^i$ = Focal loss(class_score^i, Real Classes), L_1 loss(Q_p^i, Real Boxes), Rotate IoU loss(Q_p^i, Real Boxes)
end for
return $\mathcal{L} = \sum \lambda_{cls}\mathcal{L}_{cls}^i + \lambda_{L_1}\mathcal{L}_1^i + \lambda_{iou}\mathcal{L}_{iou}^i$

IV. Experiment

A. Datasets

We conduct our experiments on six common datasets. *DIOR-R* [35] is a large-scale oriented object detection dataset for remote sensing images. It consists of 23 463 images and 192 512 instances that belong to 20 common classes. We train our model on the training and validation sets and test it on the test set. *DOTA* series [36] includes oriented object detection datasets for remote sensing images. They include DOTA-v1.0 [36]/ v1.5 / v2.0 [37], which differ in the number of images, instances, and categories. The images range in size from 800 × 800 to 4000 × 4000 pixels and cover various scenes and objects. The categories of *DOTA-v1.0* are 15. It contains 2806 images and 188 282 instances. The categories of *DOTA-v1.5* are 16. It uses the same images as DOTA-v1.0 but adds more small instances, resulting in 403 318 instances in total. The categories of *DOTA-v2.0* are 18. It contains 11 268 images and 1 793 658 instances. We train our model on the training and validation sets of these datasets and test it on the test set. We submit test results to the official evaluation server of DOTA to obtain the detection performance. *HRSC2016* [38] is a challenging dataset for ship detection in remote sensing images, including 1061 images. The dataset is divided into two sets: training and testing with 617 and 444 images, respectively. We evaluate our model on the test set using two metrics, the PASCAL VOC07 and

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TABLE I
EXPERIMENT SETTINGS OF ORIENTEDFORMER

Method	config	value
OrientedFormer	optimizer	AdamW
	base learning rate	5e-5
	weight decay	1e-6
	optimizer momentum	β_1, β_2=0.9, 0.999
	batch size	4
	GPUs	2
	epochs	12 or 24
	lr decay epochs	(8, 11) or (16, 22)
	warmup iter	500
	warmup factor	0.333
	clip gradient type	full model
	clip gradient value	1.0
	clip gradiant norm	2.0
	data augmentation	RandomFlip
	seed	Random Seed

VOC12. *ICDAR2015* [39] is utilized for text detection and comprises 1000 training images along with 500 test images.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: We conduct all the experiments on two NVIDIA RTX 2080ti with a batch of four (two images per GPU). Models are constructed based on MMRotate [48] with PyTorch. The ResNet [30], Swin [49], and LSK [50] are used as backbone pre-trained on the ImageNet [51]. We optimize models with AdamW optimizer [52] with learning rate $5 \times e^{-5}$. The weights of losses are 5.0, 2.0, and 2.0 for rotated IoU loss [34], focal loss [33], and L1 loss, respectively. Data augmentation strategies contain only random flips. Details of experiments are displayed in Table I and Algorithm 1.

For experiments on the DOTA-v1.0/1.5/2.0, images are cropped into patches of 1024 × 1024 with overlaps of 200 and trained for 12 epochs. At epochs 8 and 11, the learning rate is divided by 10. In addition, for the multiscale training in DOTA-v1.0, images are first resized at three scales (0.5, 1.0, and 1.5) and then cropped following single-scale training. For experiments on DIOR-R, images are trained for 12 epochs with the original fixed size of 800 × 800. For experiments on HRSC2016, we scale images to a range of (512, 800) and train them for 24 epochs. Images on ICDAR2015 are trained for 24 epochs with a fixed size of 800 × 800.

2) Evaluation Metrics: AP_{50}, AP_{75}, and $AP_{50:95}$ measure the accuracy of methods. We also analyze precision, recall, F-measure, PASCAL VOC07 and VOC12 metrics for different methods. FPS is a metric for assessing inference speed. Params and FLOPs are used to count the parameters and complexity, respectively, of the model. Epochs are used to measure model training time.

C. Comparisons With State-of-the-Arts

1) Results on DIOR-R: We compare OrientedFormer with modern CNN-based and transformer-based detectors. The detailed results of every category on the DIOR-R [35] are reported in Table II and Figs. 6 and 7. Results of compared methods are from their papers. Our method achieves 65.07% AP_{50} with LSK-T, 67.28% AP_{50} with ResNet50, and 68.84% AP_{50} with Swin-T, outperforming all comparison CNN-based one-stage and two-stage detectors and transformer-based detectors.

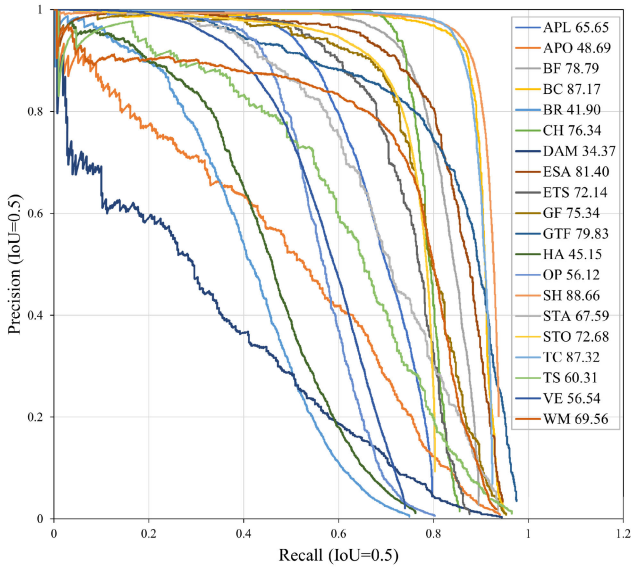


Fig. 6. *P–R curve (IoU = 0.5)* on the DIOR-R test set with ResNet50.

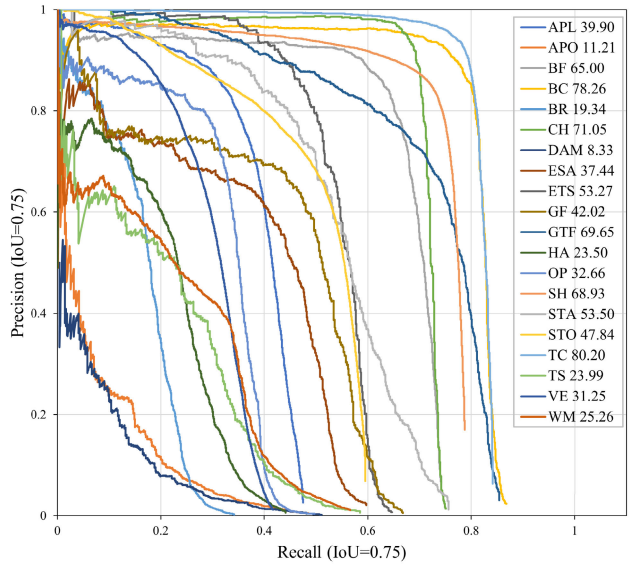


Fig. 7. *P–R curve (IoU = 0.75)* on the DIOR-R test set with ResNet50.

2) Results on DOTA-v1.0: We report results on DOTA-v1.0 in Table III, compared with current CNN-based and transformer-based detectors. Results of compared methods are from their papers. In terms of accuracy measured by AP_{50}, OrientedFormer achieves 75.37% AP_{50} with ResNet50, 75.88% AP_{50} with Swin-T, and 75.92% AP_{50} with ResNet101 under single-scale data. Additionally, it achieves 79.06% AP_{50} with ResNet50 under multiscale data.

3) Results on DOTA-v1.5: Table IV shows a comparison of our method with other modern detectors, using results from their papers. Using the ResNet50 backbone, our method achieves 67.06% AP_{50} with single-scale data. DOTA-v1.5 contains many small object instances, for example, small vehicle

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TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE *DIOR-R*. THE RESULTS IN **BOLD** DENOTE THE BEST PERFORMANCE OF EACH COLUMN

Method	Backbone	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	AP_{50}
one-stage:																						
RetinaNet-O [33]	R50	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
DFDet [40]	R50	61.92	38.83	77.41	81.36	34.11	74.97	26.26	62.31	76.06	75.56	79.62	38.26	52.76	80.40	73.11	68.27	81.38	52.23	44.11	63.35	62.11
Oriented Rep [9]	R50	**70.03**	46.11	76.12	87.19	39.14	**78.76**	34.57	71.80	80.42	76.16	79.41	45.48	54.90	87.82	77.03	68.07	81.60	56.83	51.57	**71.25**	66.71
DCFL [7]	R50	68.60	53.10	76.70	87.10	42.10	78.60	34.50	71.50	80.80	**79.70**	79.50	47.30	57.40	85.20	64.60	66.40	81.50	58.90	50.90	70.90	66.80
two-stage:																						
Gliding Vertex [6]	R50	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
RoI Transformer [41]	R50	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	**78.23**	70.26	81.61	54.86	43.27	65.52	63.87
QPDet [42]	R50	63.22	41.39	71.97	**88.55**	41.23	72.63	28.82	**78.90**	69.00	70.07	**83.01**	47.83	55.54	81.23	72.15	62.66	**89.05**	58.09	43.38	65.36	64.20
AOPG [35]	R50	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99	64.41
DODet [43]	R50	63.40	43.35	72.11	81.32	**43.12**	72.59	33.32	78.77	70.84	74.15	75.47	**48.00**	**59.31**	85.41	74.04	71.56	81.52	55.47	51.86	66.40	65.10
end-to-end:																						
ARS-DETR [20]	R50	68.00	54.17	74.43	81.65	41.13	75.66	34.89	73.07	81.92	76.10	78.62	36.33	55.41	84.55	70.09	72.23	81.14	61.52	50.57	70.28	66.12
end-to-end:																						
OrientedFormer	LSK-T	58.89	42.64	78.56	84.56	37.49	74.05	33.39	71.97	79.25	74.33	80.73	43.10	52.84	87.89	66.08	68.85	86.61	58.30	55.48	66.31	65.07
OrientedFormer	R50	65.65	48.69	**78.79**	87.17	41.90	76.34	34.37	72.14	81.40	75.34	79.83	45.15	56.12	**88.66**	67.59	72.68	87.32	60.31	56.54	69.56	67.28
OrientedFormer	Swin-T	67.45	**50.81**	78.47	86.19	42.92	77.88	**40.76**	75.52	**84.13**	77.86	81.76	46.19	56.53	88.64	75.32	**73.81**	86.72	**60.80**	**56.88**	68.09	**68.84**

TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE *DOTA-v1.0* DATASET. * DENOTES MULTISCALE TRAINING AND TESTING. THE RESULTS IN **BOLD** DENOTE THE BEST PERFORMANCE OF EACH COLUMN

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	AP_{50}
one-stage:																	
PSC [17]	R50	88.24	74.42	48.63	63.44	79.98	80.76	87.59	90.88	82.02	71.58	59.12	60.78	65.78	71.21	53.06	71.83
R3Det [15]	R101	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
S^2A-Net [16]	R50	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
H2RBox [44]	R50	88.93	78.89	46.27	68.79	81.12	75.45	86.68	90.89	86.71	**87.33**	64.15	68.83	62.81	69.39	59.79	74.40
CFA [18]	R50	88.34	83.09	51.92	72.23	79.95	78.68	87.25	90.90	85.38	85.71	59.63	63.05	73.33	70.36	47.86	74.51
DHRec [45]	R50	88.58	77.90	53.84	72.93	78.45	78.84	87.64	90.88	88.78	85.46	56.11	66.74	67.58	70.25	57.53	74.57
DFDet [40]	R50	88.92	79.25	48.40	70.00	80.22	78.85	87.21	90.90	83.13	83.98	60.07	66.49	68.27	76.78	58.11	74.71
SASM [8]	R50	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
two-stage:																	
RoI Transformer [41]	R50	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	52.81	53.95	67.61	74.67	68.75	61.03	74.61
Gliding Vertex [6]	R101	89.64	**85.00**	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
AOPG [35]	R50	89.27	83.49	52.50	69.97	73.51	82.31	87.95	90.89	87.64	84.71	60.01	66.12	74.19	68.30	57.80	75.24
DODet [43]	R50	89.34	84.31	51.39	71.04	79.04	82.86	88.15	90.90	86.88	84.91	62.69	67.63	75.47	72.22	45.54	75.49
Oriented R-CNN [4]	R50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
SCRDet++ * [46]	R101	**89.77**	83.90	56.30	73.98	72.60	75.63	82.82	90.76	**87.89**	86.14	65.24	63.17	76.05	68.06	70.24	76.20
end-to-end:																	
D. DETR-O [12]	R50	84.89	70.71	46.04	61.92	73.99	78.83	87.71	90.07	77.97	78.41	47.07	54.48	66.87	67.66	55.62	69.48
D. DETR-O w/ CSL [47]	R50	86.27	76.66	46.64	65.29	76.80	76.32	87.74	90.77	79.38	82.36	54.00	61.47	66.05	70.46	61.97	72.15
ARS-DETR [20]	R50	86.97	75.56	48.32	69.20	77.92	77.94	87.69	90.50	77.31	82.86	60.28	64.58	74.88	71.76	66.62	74.16
end-to-end:																	
OrientedFormer	R50	88.14	79.13	51.96	67.34	81.02	83.26	88.29	**90.90**	85.57	86.25	60.84	66.36	73.81	71.23	56.49	75.37
OrientedFormer	Swin-T	88.74	78.94	53.43	72.05	81.08	84.22	88.40	90.90	86.23	86.65	61.05	63.11	75.78	73.02	54.62	75.88
OrientedFormer	R101	88.18	82.14	52.60	72.00	80.88	83.11	88.35	90.87	84.08	86.31	63.18	67.26	76.58	69.08	54.12	75.92
OrientedFormer*	R50	87.92	83.29	**58.92**	**80.90**	**81.93**	**84.62**	**88.81**	90.89	86.81	86.95	**66.68**	59.86	**78.82**	**77.88**	**71.68**	**79.06**

TABLE IV

MAIN RESULTS OF SMALL SIZE OBJECTS ON *DOTA-v1.5*. THE RESULTS IN **BOLD** DENOTE THE BEST PERFORMANCE OF EACH COLUMN

Method	SV	LV	SH	ST	SP	CC	AP_{50}
RetinaNet-O [33]	44.53	56.79	73.31	59.96	64.52	0.83	59.16
Faster RCNN-O [53]	51.28	68.98	79.37	67.50	65.28	1.54	62.00
Mask R-CNN [54]	51.31	71.34	79.75	66.07	64.46	9.42	62.67
HTC [55]	51.54	73.31	80.31	67.34	64.48	5.15	63.40
ReDet [5]	52.38	75.73	80.92	68.64	70.55	**11.53**	66.86
OrientedFormer	**64.05**	**77.04**	**85.33**	**78.11**	**72.08**	10.86	**67.06**

(SV), ship (SH), and swimming pool (SP). For these instances, our method performs better.

4) Results on ICDAR2015: We conduct experiments using the ICDAR2015 [39], shown in Table V. We reimplement compared methods using MMRotate, with the same settings as our methods. Our OrientedFormer achieves a precision of 85.3%, a recall of 74.2%, and an F-measure of 79.4%.

5) Results on HRSC2016: The HRSC2016 contains only ships. Table VI shows the results of our method and other object detectors from their papers. Our OrientedFormer achieves 90.17% and 96.48% AP_{50} with ResNet50 under VOC07 and VOC12 metrics, respectively, which are competitive with modern detectors.

6) Results on DOTA-v2.0: As shown in Table VII, our proposed OrientedFormer is compared with CNN-based one-stage and two-stage detectors, using results reported in their papers. For a fair comparison, the backbones of all models are ResNet50. Our method achieves the state-of-the-art performance of 54.27% AP_{50} on the DOTA-v2.0 benchmark with

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TABLE V

MAIN RESULTS OF PRECISION, RECALL (IOU = 0.5), F-MEASURE, AND FLOPS ON *ICDAR2015*. THE RESULTS IN **BOLD** DENOTE THE BEST PERFORMANCE. THE BACKBONE USED BY ALL METHODS IS RESNET50

Method	Precision	Recall	F-measure	FLOPs
SASM [8]	56.7	77.9	65.7	71G
PSC [17]	83.7	63.2	72.0	78G
Retinanet-O [33]	83.9	67.5	74.8	77G
GWD [56]	84.4	67.6	75.1	77G
R3DET [15]	83.2	69.2	75.6	120G
Oriented RCNN [4]	73.9	**80.9**	77.2	100G
CFA [18]	78.3	77.1	77.7	71G
Oriented Reppoints [9]	79.0	77.0	78.0	71G
S^2A-Net [16]	81.6	75.6	78.5	77G
RoI Transformer [41]	78.2	80.7	79.4	55G
OrientedFormer	**85.3**	74.2	**79.4**	195G

TABLE VI

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE *HRSC2016* DATASET. 07 MEANS EVALUATION UNDER PASCAL VOC2007 METRIC, AND 12 MEANS EVALUATION UNDER PASCAL VOC2012 METRIC

Methods	Backbone	mAP(07)	mAP(12)
PSC [17]	R50	85.65	-
RoI Transformer [41]	R101	86.20	-
Gliding Vertex [6]	R101	88.20	-
PIoU [57]	DLA34	89.20	-
CenterMap [58]	R50	-	92.8
R3Det [15]	R101	89.26	96.01
CSL [47]	R101	89.62	96.10
H2RBox-v2 [59]	R50	89.66	-
GWD [56]	R101	89.85	97.37
OSKDet [60]	R101	89.98	-
S^2A-Net [16]	R101	90.17	95.01
Oriented R-CNN [4]	R50	90.40	96.50
OrientedFormer	R50	90.17	96.48

TABLE VII

PERFORMANCE COMPARISONS ON THE *DOTA-v2.0* DATASET

Method				
AP_{50}	SASM [8] 44.53	RetinaNet-O [33] 46.68	Oriented Rep [9] 48.95	Mask R-CNN [54] 49.47
Method				
AP_{50}	ATSS-O [61] 49.57	S^2A-Net [16] 49.86	HTC [55] 50.34	DCFL [7] 51.57
Method				
AP_{50}	RoI Trans. [41] 52.81	S^2A-Net+DCFL 52.84	Oriented R-CNN [4] 53.28	**OrientedFormer 54.27**

single-scale data. Our method outperforms all compared CNN-based detectors.

D. Ablation Study

1) Numbers of Object Queries: In this experiment, we evaluate the impact of the number of object queries, as illustrated in Table VIII(a). We noticed a significant improvement in performance as the number of object queries increased. With 100 object queries, the AP_{50} is only 65.16%, but when the number of object queries is increased to 300, it rises to 67.28% (an improvement of 2.12%). This suggests that a sufficient number of object queries can effectively cover the objects in images.

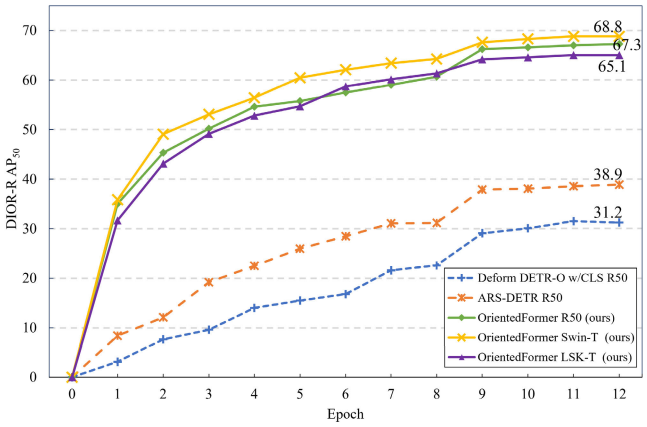


Fig. 8. *Convergence curves* of Deformable DETR-O with CLS, ARS-DETR, OrientedFormer with ResNet50, Swin-T, and LSK-T on DIOR-R.

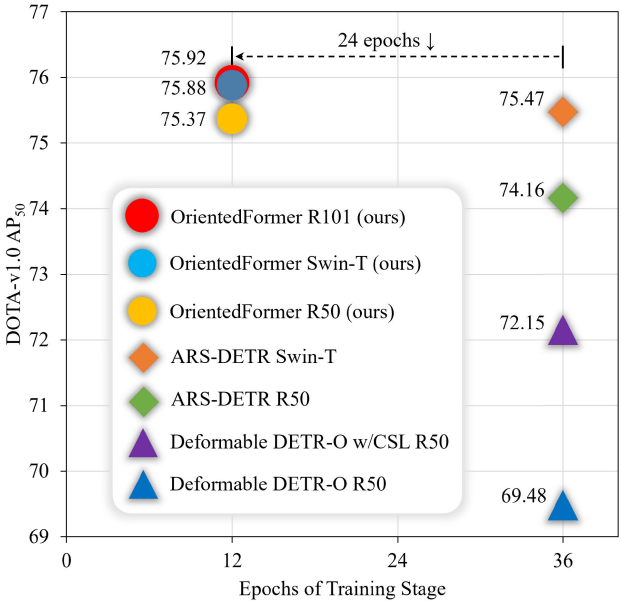


Fig. 9. *Epochs* of training stage versus *accuracy* on the DOTA-v1.0 test set.

2) Number of Sampling Points in Oriented Cross-Attention: In this ablation study, we use a different number of sampling points, shown in Table VIII(b). The reason why numerous sampling points are used is that features sampled by these sampling points are responsible for the classification and regression of objects, and spatial-aware attention mainly focuses on these features. As we increase the number of sampling points from 4 to 32, the AP_{50} grows from 65.60% to 67.28%. It indicates that abundant features facilitate spatial-aware attention and the entire decoder.

3) Number of Attention Heads in Oriented Cross-Attention: In this experiment, we use different numbers of attention heads, shown in Table VIII(c). The reason why it is necessary to use multiple heads is that different heads establish different associations between queries and values. As we increase the number of heads from 8 to 64, the AP_{50} grows from 66.33% to 67.28%. It indicates that heads in attention can provide multiple subspaces for representation, and extends the ability to focus on different parts of features.

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TABLE VIII

ORIENTEDFORMER ABLATION EXPERIMENTS WITH RESNET-50 ON DIOR-R. DEFAULT CHOICE FOR OUR MODEL IS COLORED GRAY. (a) ABLATION OF DIFFERENT NUMBER OF *Object Queries*. (b) ABLATION OF DIFFERENT NUMBER OF *Sampling Points* PER HEAD IN ORIENTED CROSS-ATTENTION. (c) ABLATION OF DIFFERENT NUMBER OF *Heads* IN ORIENTED CROSS-ATTENTION

Queries	100	200	**300**	400	500
AP_{50}	65.16	66.42	**67.28**	67.13	66.93

(a) The ablation of different number of **object queries**.

Points	8	16	**32**	64	128
AP_{50}	65.60	67.02	**67.28**	66.27	66.97

(b) The ablation of different number of **sampling points** per head in oriented cross-attention.

Heads	8	16	32	**64**	128
AP_{50}	66.33	66.63	66.73	**67.28**	66.78

(c) The ablation of different number of **heads** in oriented cross-attention.

TABLE IX

COMPARISONS OF DIFFERENT *PE* IN THE DECODER

PE	-	Deform.	DAB.	Learnable	Gaussian
AP_{50}	66.85	65.97	66.03	64.27	**67.28**

TABLE X

COMPARISONS OF DIFFERENT *Self-Attention* IN THE DECODER

Self-Attention	-	iof	iou	Wasserstein
AP_{50}	67.03	66.57	67.08	**67.28**

TABLE XI

EFFECTIVENESS OF PROPOSED *Individual Modules* ON DIOR-R

Methods	Gaussian PE	Wasserstein Self-Attention	Oriented Cross-Attention	DIOR-R		
				AP_{50}	AP_{75}	$AP_{50:95}$
Oriented Former				62.69	44.18	41.38
	✓	✓		63.08	43.44	41.00
			✓	65.78	43.69	41.87
		✓	✓	66.85	46.22	43.73
	✓		✓	67.03	44.07	42.49
	✓	✓	✓	67.28	44.13	42.66

TABLE XII

EFFECTIVENESS OF *Individual Modules* ON DOTA-V1.0

Methods	Gaussian PE	Wasserstein Self-Attention	Oriented Cross-Attention	DOTA-v1.0		
				AP_{50}	AP_{75}	$AP_{50:95}$
Oriented Former				73.81	47.74	45.40
	✓	✓		74.55	49.26	46.28
			✓	74.64	47.80	45.85
		✓	✓	74.69	46.16	45.12
	✓		✓	74.76	48.95	45.97
	✓	✓	✓	75.37	46.39	45.01

E. Comparisons of Different PEs

We perform ablation experiments with different PE, shown in Table IX. The learnable PE [10], sinusoidal absolute PE following Deformable DETR [12], and DAB DETR [28] are compared. They only encode sizes and positions of oriented boxes, but lack angles. The model achieves 66.85% without any PE, due to a lack of information of sequence ordering. When the PE of Deformable DETR and DAB DETR and the learnable one are used, the AP_{50} is reduced. We argue that these PEs are mismatched for positional queries. Our Gaussian PE can bring an improvement in performance.

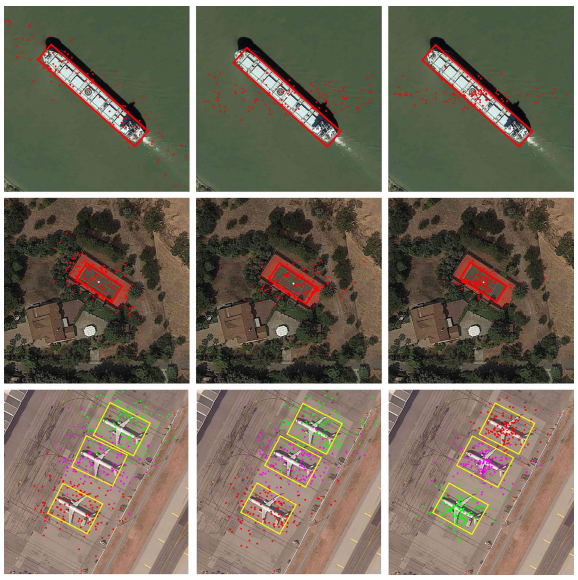


Fig. 10. Sampling points in cross-attention. Oriented cross-attention (left column) rotates sampling points by angles for alignment. Deformable offsets method (middle column) does not rotate sampling points. Random offsets method (right column) employs random sampling points.

F. Comparisons of Different Self-Attention

In Table X, we compare Wasserstein self-attention with other modern self-attention. When vanilla self-attention [10], [12] is applied, the model only achieves an AP_{50} of 67.03%. We replace Gaussian Wasserstein distance scores with Intersection over Foreground (IoF) and Intersection over Union (IoU) for comparison. Compared with IoF and IoU methods, our proposed Wasserstein self-attention yields the best performance, achieving improvements of 0.71% and 0.2%, respectively.

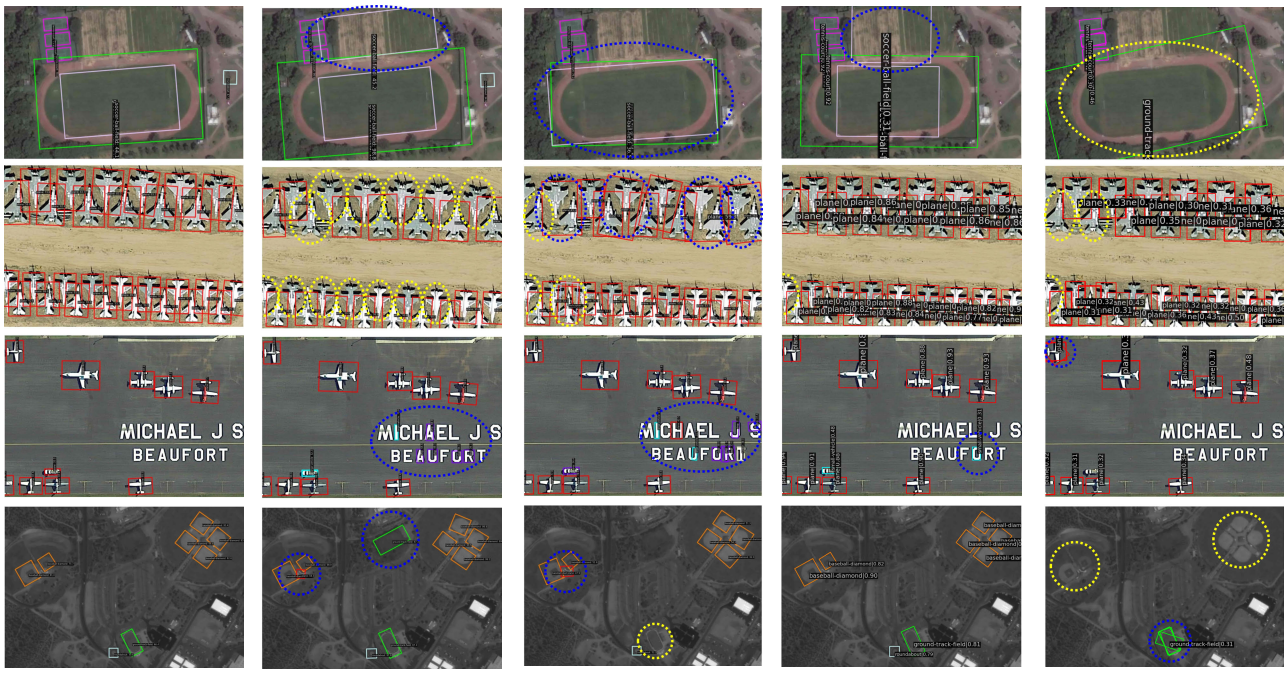
G. Effects of Proposed Individual Strategy

In this study, we evaluate the effectiveness of each strategy proposed in our method, including Gaussian PE, Wasserstein self-attention, and oriented cross-attention, as shown in Tables XI and XII. Incrementally incorporating each individual strategies, they improve performance.

H. Convergence and Training Epochs

DETR [10] suffers from slow convergence and long training times. To further investigate convergence, we compare OrientedFormer with other end-to-end models, depicted in Fig. 8.

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(a) (b) (c) (d) (e)

Fig. 11. Comparison between our method and others on challenging samples. Confidence threshold is set to 0.3. Blue circles denote error results. Yellow circles denote missed results. First line: large-scale objects, for example, soccer fields and ground track fields. Second line: densely packed objects, for example, planes. Third line: images with complex backgrounds, for example, the disturbance of letters. Fourth line: images under poor environmental conditions. (a) OrientedFormer (ours). (b) Oriented R-CNN [4]. (c) SASM [8]. (d) ARS-DETR [20]. (e) Deformable DETR-O [12].

For fair comparisons, all methods are trained on 12 epochs on DIOR-R with 300 queries. OrientedFormer achieves an AP_{50} of 67.3% in just 12 epochs with ResNet50, surpassing Deformable DETR-O with CSL (31.2%) and ARS-DETR (38.9%).

We compare OrientedFormer with other end-to-end models for accuracy and training epochs, shown in Fig. 9. Training OrientedFormer for 12 epochs can outperform ARS-DETR and Deformable DETR-O, which require 36 epochs of training. Specifically, OrientedFormer achieves an AP_{50} of 75.37% with ResNet50 in 12 epochs, while ARS-DETR achieves only 74.16% and Deformable DETR-O achieves only 69.48% in 36 epochs.

I. Comparison of Speed, Parameters, FLOPs, and Accuracy

To further explore the performance of OrientedFormer, we conducted an experiment comparing its FPS, parameters, FLOPs, and AP_{50} with other methods on DOTA-v1.0, shown in Table XIII. OrientedFormer outpaces CNN-based two-stage methods and other end-to-end methods in speed but slightly trails behind most one-stage methods. Additionally, OrientedFormer has a slightly larger number of parameters compared to other end-to-end methods.

J. Comparison of Different Feature Layers of Backbone

Backbones extract features from images and play an important role in oriented object detection. We conducted an experiment with different feature layers of backbone ResNet50, shown in Table XIV. As the number of backbone network feature layers increases, the AP, parameters, and FLOPs of the model gradually increase. Four layers of features are selected because multiscale features can capture rich information from the images.



Fig. 12. Center points of learned positional queries.

K. Comparison of Different Sampling Methods

Learnable sampling points P are rotated by angles of positional queries for alignment in our oriented cross-attention. We compare our methods with different sampling strategies, shown in Table XV. The fixed offset method utilizes fixed

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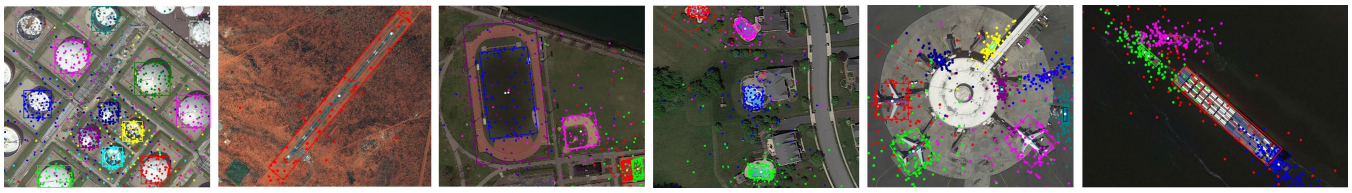


Fig. 13. Visualization of oriented cross-attention. For readability, we draw the center of positional queries and sampling points.

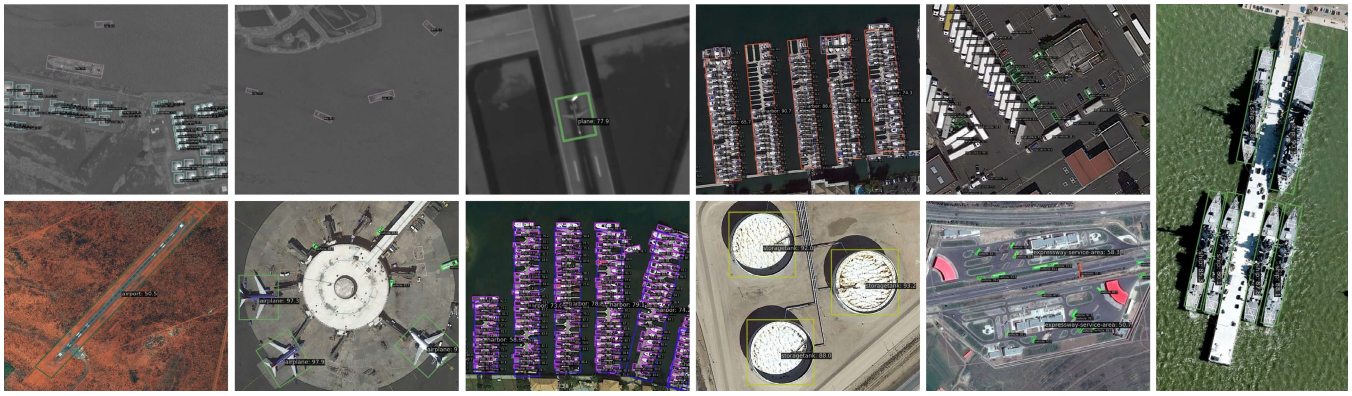


Fig. 14. Visualization results of our method for DOTA, DIOR, and HRSC2016. DOTA contains images depicting extreme weather and poor lighting conditions. Oriented boxes, labels, and confidences are drawn.

TABLE XIII
SPEED, PARAMETERS, FLOPS, AND ACCURACY ON DOTA-V1.0

Method	Frame	Backbone	FPS	Params	FLOPs	AP_{50}
RoI Transformer [41]	Two-stage	R50	9.2	55M	253G	74.61
Oriented RCNN [4]		R50	7.3	41M	225G	75.87
Gliding Vertex [6]		R101	10.2	41M	225G	75.02
R3Det [15]	One-stage	R101	6.1	42M	335G	73.79
CFA [18]		R50	16.6	37M	194G	74.51
SASM [8]		R50	15.8	37M	194G	74.92
PSC [17]		R50	16.8	37M	218G	71.83
DCFL [7]		R50	17.7	36M	216G	74.26
D.DETR-O [12]	End-to-end	R50	10.8	41M	186G	69.48
D.DETR-O w/CSL [47]		R50	10.7	41M	186G	72.15
ARS-DETR [20]		R50	10.1	42M	186G	74.16
OrientedFormer	End-to-end	R50	11.4	44M	529G	75.37
OrientedFormer		Swin-T	11.2	45M	536G	75.88
OrientedFormer		R101	11.1	63M	606G	75.92

TABLE XIV
COMPARISONS OF *Different Layers* OF BACKBONE ON DIOR-R

Method	Backbone	Layers	AP_{50}	AP_{75}	$AP_{50:95}$	Params	FLOPs
Oriented -Former	R50	1	56.21	33.54	33.23	41M	287G
		2	60.69	39.86	38.35	41M	297G
		3	66.37	44.14	42.35	42M	315G
		4	67.28	44.13	42.66	44M	325G

sampling points around the center points of positional queries. The deformable offsets method does not rotate sampling points, which is the main difference between our methods. The random offsets method employs random sampling points around the center points of positional queries. We visualize sampling points of different methods in Fig. 10. Our oriented cross-attention can more effectively align features and thus focuses on more accurate object features.

TABLE XV
COMPARISON OF *Different Sampling* METHODS ON DIOR-R

	Methods	AP_{50}	AP_{75}
OrientedFormer	Fixed Offsets	65.98	43.02
	Deformable Offsets	66.66	44.10
	Random Offsets	66.70	43.47
	Oriented Cross-attention	**67.28**	**44.13**

L. Visualization

1) Comparison With Other Methods: We compare our method with others on large-scale objects, densely packed objects, images with complex backgrounds, and images under low lighting conditions, shown in Fig. 11. Other methods struggle with accurately detecting large objects, often miss densely packed objects, mistake background noise for objects, and fail to perform well under poor environmental conditions.

2) Learnable Positional Queries: Positional queries are used for locations of objects. We visualize the center points of learned positional queries shown in Fig. 12. Default 300 positional queries are used in the experiments. Positional queries are positioned at the center of objects and suspected objects. It demonstrates the utility of positional queries for representing object positions.

3) Learnable Sampling Points: For a better understanding of the learned oriented cross-attention, we visualize centers of positional queries and sampling points in the decoder, as shown in Fig. 13. For readability, all sampling points are scaled to original images. Features sampled from sampling points play the role of values in cross-attention. Sampling

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Fig. 15. Suboptimal results.

points have aligned with oriented boxes. Most of the sampling points are distributed within oriented boxes and others out of boxes.

4) Detection Results: We visualize detection results on different datasets. Fig. 14 shows the detection results on DOTA, DIOR-R, and HRSC2016. Oriented boxes locate objects accurately in images. It is worth noting that the DOTA dataset contains numerous images depicting extreme weather and poor lighting conditions. However, our method remains effective for detection under these circumstances.

5) Suboptimal Results: We show some suboptimal results in Fig. 15. There are a large number of objects with small size and huge aspect ratios in oriented remote sensing images. In addition, some foreground objects are similar to the background. These challenges warrant further investigation in future research.

V. Conclusion

In this article, we propose an end-to-end transformer-based detector OrientedFormer for oriented object detection in remote sensing images. The proposed OrientedFormer comprises the Gaussian PE, Wasserstein self-attention, and oriented cross-attention. These dedicated components work together to accurately classify and localize objects with multiple orientations in remote sensing images. First, Gaussian PE is introduced to encode not only the position and size but also the angles of oriented boxes. Second, Wasserstein self-attention is proposed to incorporate geometric relations between content queries into the self-attention mechanism. Finally, oriented cross-attention is designed to address misalignment issues by rotating sampling points according to object angles. Extensive experiments on six datasets demonstrate the effectiveness of our methods. We validate that transformer-based detectors can be competitive with CNN-based one-stage and two-stage detectors. In comparison to previous end-to-end detectors, the OrientedFormer achieves a performance increase of 1.16 and 1.21 AP_{50} on DIOR-R and DOTA-v1.0, respectively, while also reducing training epochs from $3\times$ to $1\times$.

Limitations: We defer the task of further reducing the number of parameters and enhancing the inference speed of the model to future research endeavors. In oriented object detection, the rotation-equivariant network is sensitive to the orientations of objects. In the future, we hope to construct an end-to-end transformer-based rotation-equivariant-oriented object detector.

References

[1] D. Biswas and J. Tešic, "Small object difficulty (SOD) modeling for objects detection in satellite images," in *Proc. 14th Int. Conf. Comput. Intell. Commun. Netw. (CICN)*, Dec. 2022, pp. 125–130.

[2] D. Biswas, M. M. M. Rahman, Z. Zong, and J. Tesic, "Improving the energy efficiency of real-time DNN object detection via compression, transfer learning, and scale prediction," in *Proc. IEEE Int. Conf. Netw., Archit. Storage (NAS)*, Oct. 2022, pp. 1–8.

[3] D. Biswas and J. Tešić, "Unsupervised domain adaptation with debiased contrastive learning and support-set guided pseudolabeling for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3197–3210, 2024.

[4] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[5] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[6] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2020.

[7] C. Xu et al., "Dynamic coarse-to-fine learning for oriented tiny object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 7318–7328.

[8] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 1, pp. 923–932.

[9] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented reppoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1829–1838.

[10] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[11] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[12] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–16.

[13] Z. Gao, L. Wang, B. Han, and S. Guo, "AdaMixer: A fast-converging query-based object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5354–5363.

[14] L. Dai, H. Liu, H. Tang, Z. Wu, and P. Song, "AO^2-DETR: Arbitrary-oriented object detection transformer," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2342–2356, May 2023.

[15] X. Yang, J. Yan, Z. Feng, and T. He, "R^3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.

[16] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[17] Y. Yu and F. Da, "Phase-shifting coder: Predicting accurate orientation in oriented object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13354–13363.

[18] Z. Guo, C. Liu, X. Zhang, J. Jiao, X. Ji, and Q. Ye, "Beyond bounding-box: Convex-hull feature adaptation for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8792–8801.

[19] Y. Zhou, S. Chen, J. Zhao, R. Yao, Y. Xue, and A. E. Saddik, "CLT-Det: Correlation learning based on transformer for detecting dense objects in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708915.

[20] Y. Zeng, Y. Chen, X. Yang, Q. Li, and J. Yan, "ARS-DETR: Aspect ratio-sensitive detection transformer for aerial oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610315.

[21] S. Feng and B. Wang, "PSD-SQ: Point set decoding based on semantic query for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5401412, doi: 10.1109/TGRS.2024.3352011.

[22] Q. Zhou, C. Yu, Z. Wang, and F. Wang, "D^2Q-DETR: Decoupling and dynamic queries for oriented object detection with transformers," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[23] Z. Hu et al., "EMO2-DETR: Efficient-matching oriented object detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616814, doi: 10.1109/TGRS.2023.3300154.

[24] Y. Wang, X. Zhang, T. Yang, and J. Sun, "Anchor DETR: Query design for transformer-based detector," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 3, Jun. 2022, pp. 2567–2575.

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[25] P. Gao, M. Zheng, X. Wang, J. Dai, and H. Li, "Fast convergence of DETR with spatially modulated co-attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 3621–3630.
[26] X. Dai, Y. Chen, J. Yang, P. Zhang, L. Yuan, and L. Zhang, "Dynamic DETR: End-to-end object detection with dynamic attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2988–2997.
[27] Y. Chen, X. Dai, M. Liu, D. Chen, L. Yuan, and Z. Liu, "Dynamic convolution: Attention over convolution kernels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11030–11039.
[28] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–19.
[29] S. Zhang et al., "Dense distinct query for end-to-end object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7329–7338.
[30] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[31] J. T. Barron, B. Mildenhall, M. Tancik, P. Hedman, R. Martin-Brualla, and P. P. Srinivasan, "Mip-NeRF: A multiscale representation for anti-aliasing neural radiance fields," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5855–5864.
[32] H. W. Kuhn, "The Hungarian method for the assignment problem," *Nav. Res. Logistics Quart.*, vol. 2, pp. 83–97, Mar. 1955.
[33] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[34] D. Zhou et al., "IoU loss for 2D/3D object detection," in *Proc. Int. Conf. 3D Vis. (3DV)*, Sep. 2019, pp. 85–94.
[35] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.
[36] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[37] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022.
[38] Z. Liu, H. Wang, L. Weng, and Y. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.
[39] D. Karatzas et al., "ICDAR 2015 competition on robust reading," in *Proc. 13th Int. Conf. Document Anal. Recognit. (ICDAR)*, Aug. 2015, pp. 1156–1160.
[40] X. Xie, G. Cheng, C. Rao, C. Lang, and J. Han, "Oriented object detection via contextual dependence mining and penalty-incentive allocation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5618010.
[41] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.
[42] Y. Yao et al., "On improving bounding box representations for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600111, doi: 10.1109/TGRS.2022.3231340.
[43] G. Cheng et al., "Dual-aligned oriented detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618111.
[44] X. Yang, G. Zhang, W. Li, Y. Zhou, X. Wang, and J. Yan, "H2RBox: Horizontal box annotation is all you need for oriented object detection," in *Proc. 11th Int. Conf. Learn. Represent.*, 2022, pp. 1–14.
[45] G. Nie and H. Huang, "Multi-oriented object detection in aerial images with double horizontal rectangles," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4932–4944, Apr. 2023.
[46] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2384–2399, Feb. 2023.
[47] X. Yang and J. Yan, "Arbitrary-oriented object detection with circular smooth label," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 677–694.
[48] Y. Zhou et al., "MMRotate: A rotated object detection benchmark using PyTorch," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 7331–7334.
[49] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[50] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16794–16805.
[51] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–18.
[53] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.
[54] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.
[55] K. Chen et al., "Hybrid task cascade for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4974–4983.
[56] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11830–11841.
[57] Z. Chen et al., "PIoU loss: Towards accurate oriented object detection in complex environments," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 195–211.
[58] J. Wang, W. Yang, H.-C. Li, H. Zhang, and G.-S. Xia, "Learning center probability map for detecting objects in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4307–4323, May 2021.
[59] Y. Yu, X. Yang, Q. Li, Y. Zhou, F. Da, and J. Yan, "H2RBox-v2: Incorporating symmetry for boosting horizontal box supervised oriented object detection," in *Proc. 37th Conf. Neural Inf. Process. Syst.*, 2023, pp. 1–14.
[60] D. Lu, D. Li, Y. Li, and S. Wang, "OSKDet: Orientation-sensitive keypoint localization for rotated object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1182–1192.
[61] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.



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